

7 (a) (i) The photoelectric effect is the emission of (photo)electrons from a cold metal surface when electromagnetic radiation of sufficiently high frequency is incident upon it. **B1**

(ii) Electromagnetic radiation can exist as discrete quanta of energy known as photons with energy $E = hf$ where h is the Planck constant and f is the frequency of the electromagnetic radiation. **B1**

When photons are incident on the metal, electrons in the metal can absorb and gain the photon's energy to overcome the energy needed to bring it to the surface and the work function in order to escape from the metal surface. **B1**

There is a minimum energy of photon needed to minimally overcome the work function of the metal, which corresponds to the surface-most electrons being ejected. Since the energy of the photon is directly proportional to the frequency of electromagnetic radiation, there is a minimum threshold frequency for the photoelectric effect to occur. **B1**

(b) (i) $E = hf$ **C1**

$$\begin{aligned} \text{work function} &= 6.63 \times 10^{-34} \times 8.8 \times 10^{14} \\ &= 5.8 \times 10^{-19} \text{ J} \end{aligned} \quad \textbf{A1}$$

(ii) $hf = \phi + \frac{1}{2}mv_{\text{max}}^2$ **C1**

$$6.63 \times 10^{-34} \times 11 \times 10^{14} = 5.8 \times 10^{-19} + \frac{1}{2} \times 9.11 \times 10^{-31} \times v_{\text{max}}^2 \quad \textbf{C1}$$

$$v_{\text{max}} = 5.7 \times 10^5 \text{ m s}^{-1} \quad \textbf{A1}$$

